**5** **(a)** E.m.f. is the **amount of energy transferred from non-electrical forms to electrical energy per unit charge** as it passes through a complete circuit while

p.d. is the **amount of electrical energy converted to other forms per unit charge** delivered between 2 points.

**(b)** **(i)** 0.01 V

**(ii)** Internal resistance reduces the terminal potential difference of the battery. Since the output power is proportional to its terminal p.d., the presence of internal resistance reduces the output power of the battery.

OR There is some power loss due to power dissipated due to the internal resistance.

**(c)** **(i)** $12.00 = I_1 (r + R)$ where $I_1 r = 0.01$ V and $I_1 R = 11.99$ V
Hence $R = 1199 r$,

**(ii)**
Also $12.00 = I_2 (r + R + 1000)$ where $I_2 R = 11.00$ V
Using $R = 1199 r$,
Hence $I_2 (1199 r) = 11.00 \text{ V} \Rightarrow I_2 r = 0.009174$ V
$1000 I_2 = 12.00 – 11.00 - 0.009174 = 0.9908$ V
$I_2 = 9.908 \times 10^{-4}$ A

From $12.00 = 9.908 \times 10^{-4} (r + 1199 r + 1000) \Rightarrow r = 9.26\ \Omega$

And $R = 11100\ \Omega$

**6** **(a)** Loss in EPE of the E- field = Gain in KE of the electron